EESSI

EUROPEAN ENVIRONMENT FOR SCIENTIFIC SOFTWARE INSTALLATIONS

4 Sept 2025

https://github.com/EESSI/meetings/wiki

Agenda

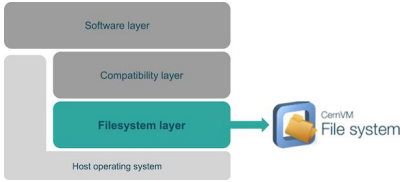
- Quick introduction by new people
- Progress update per EESSI layer
- Update on EESSI production repository `software.eessi.io`
- Update on EESSI test suite, build-and-deploy bot, documentation
- AWS/Azure sponsorship update
- EESSI governance
- Upcoming/recent events
- Q&A

Quick introduction by new people

New people on the call: feel free to introduce yourself!

- Who are you, where do you work, on what?
- Why are you interested in the EESSI project?
- Are you planning to actively contribute,
  and if so, to which aspect(s) of the project?

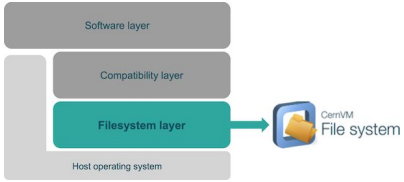
Progress update: filesystem layer



- All CernVM-FS servers have been updated to CernVM-FS 2.13.2
- New public EESSI Stratum 1 server, hosted by RAL
- Add possibility to ingest zstd-compressed tarballs #236
- **New method for ingesting tarballs** (PR #232)
  - Single staging PR to approve all builds of a particular software layer PR 🥳🎉
  - Allows for more automated checks/analyses, e.g. comparing the number of files
    - Still to be implemented

[Bob, Kenneth]

Progress update: filesystem layer



EESSIbot commented last week • edited

Please review deployment number `0` for PR1160 in EESSI/software-layer carefully.
Merging this deployment PR will lead to changes to the CernVM-FS repository `/cvmfs/software.eessi.io`.

Deployment contents

- ▼ `eessi-2025.06-software-linux-aarch64-a64fx-17562462370.tar.gz`
  - ▶ Metadata
  - ▶ Overview of payload contents
- ▶ `eessi-2025.06-software-linux-aarch64-generic-17558898880.tar.gz`
- ▶ `eessi-2025.06-software-linux-aarch64-neoverse_n1-17558881750.tar.gz`
- ▶ `eessi-2025.06-software-linux-aarch64-neoverse_v1-17558865080.tar.gz`
- ▶ `eessi-2025.06-software-linux-aarch64-nvidia-grace-17558847970.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-amd-zen2-17558944360.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-amd-zen3-17558915080.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-amd-zen4-17558895040.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-generic-17558954380.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-intel-cascadelake-17558916060.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-intel-haswell-17558929760.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-intel-icelake-17558900110.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-intel-sapphirerapids-17558920780.tar.gz`
- ▶ `eessi-2025.06-software-linux-x86_64-intel-skylake_avx512-17558912740.tar.gz`

Deployment analysis

TO BE DONE

Recommended action

TO BE DONE

Deployment contents

- ▼ `eessi-2025.06-software-linux-aarch64-a64fx-17562462370.tar.gz`
  - ▶ Metadata
  - ▼ Overview of payload contents

    Total number of items in the tarball: 22574

    URL to the tarball: https://s3.amazonaws.com/software.eessi.io-2025.06/2025.06/software/linux/aarch64/a64fx/17562462370/eessi-2025.06-software-linux-aarch64-a64fx-17562462370.tar.gz

    Summarized overview of the contents of the tarball:

```
2025.06/software/linux/aarch64/a64fx/modules/all/BLIS/1.0-GCC-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/CMake/3.29.3-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/FFTW/3.3.10-GCC-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/FlexiBLAS/3.4.4-GCC-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/OpenBLAS/0.3.27-GCC-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/OpenSSL/3.lua
2025.06/software/linux/aarch64/a64fx/modules/all/Perl/5.38.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/Perl/5.38.2-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/Python/3.12.3-GCCcore-13.3.0.lua
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2025.06/software/linux/aarch64/a64fx/modules/all/UnZip/6.0-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/cURL/8.7.1-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/gfbf/2024a.lua
2025.06/software/linux/aarch64/a64fx/modules/all/libarchive/3.7.4-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/libffi/3.4.5-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/make/4.4.1-GCCcore-13.3.0.lua
2025.06/software/linux/aarch64/a64fx/modules/all/pkgconf/1.8.0.lua
```
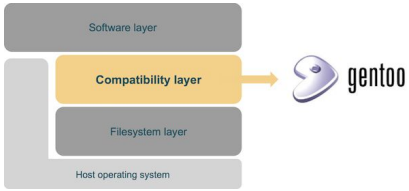

[Bob, Kenneth]

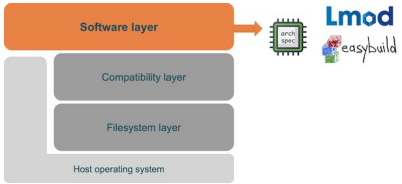
Progress update: compatibility layer



- Discussion about revising the symlink strategy for drivers
  - https://github.com/EESSI/compatibility-layer/issues/226

[Bob, Kenneth]

Progress update: software layer (1/2)



Split of EESSI/software-layer and EESSI/software-layer-scripts GitHub repositories!
See next slides on this

Highlights of recently merged software PRs (2023.06):

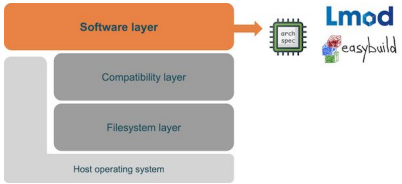
- EasyBuild 5.1.1 (PR #1135), Java 11.0.27, 17.0.15, 21.0.7 (PR #1136 & PR #1102), Bazel 6.1.0 ml_dtypes 0.3.2 tensorboard 2.15.1 (PR #1131), openCARP 17.0 (PR #1118), Qt5 dependencies (PR #1145), Octave 10.1.0 (PR #1142), scCODA 0.1.9 (PR #1128), MrBayes 3.2.7 (PR #1157), PDT 3.25.2 (PR #1171)

Highlights of recently merged software PRs (2025.06):

- GCC 13.3.0 + 14.2.0 PR #1146, gfbf/2024a + FFTW 3.3.10 PR #1160

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias, Danilo …]

Progress update: software layer (2/2)



Recent PRs for general improvements:

- Also check on EESSI 2025.06 w.r.t. missing installations and software layer being in sync across CPU targets (PR# 1162)
- Fix symlink failure easystacks (PR# 1156)

A64FX CPU target (via service account on Deucalion):

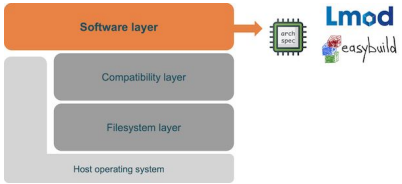
- lit 18.1.7 and astropy 7.0.0 PR #1153, BWA 0.7.18 PR #1150

Rebuilds:

- EasyBuild 5.1.1 rebuilds PR #1139 & PR #1140

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Helena …]

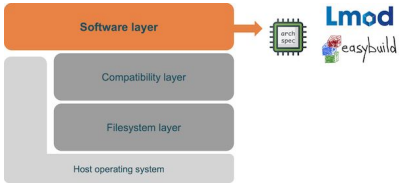
Software layer repository split



- The software-layer repository was split on 11 June 2025
- Main changes
  - Code & scripts that are used to build the EESSI software layer have been relocated to a separate repository: **EESSI/software-layer-scripts**.
  - The minimal `bot/build.sh` script in `EESSI/software-layer` pulls in the latest `main` branch of the `EESSI/software-layer-scripts` repository, symlinks the files in there, and then calls out to the `bot/build.sh` script located in that separate repository.
  - The default branch of the software-layer repository has been changed to `main` (was `2023.06-software.eessi.io`), and houses easystack files for all versions of EESSI (not just `2023.06`).
  - See more details about this change in the support portal: https://gitlab.com/eessi/support/-/issues/139

[Kenneth, Alan …]

Progress update: software-layer-scripts



Highlights of recently merged software-layer-scripts PRs:

- Add `EESSI/2025.06` module file (+ populate scripts for all CPU targets in 2025.06 software layer) PR #30
- Add hook that copies the easybuild subdir of every installation to a central reprod directory PR #24
- Allow users to configure EESSI module to be sticky PR #41; add a family name PR #43
- Use EasyBuild hooks to limit toolchains possible for specific EESSI version PR #42
- Don't use an accel subdirectory when installing CUDA and libraries to unbreak symlinks PR #46
- Create utils function for `nvidia-smi` check PR #22
- Add settings for EasyBuild 5+ to `EESSI-extend` PR #45
- Make `EESSI-extend` support accelerator installations PR #27
- Add EESSI version 2025.06 to CI workflows #40

+ various other minor PRs: #2, #22, #28, #29, #31, #32, #33, #34, #36, #38, #39, #46, #50, #52, #54, #57, #59, #67, #72

[Kenneth, Alan, …]

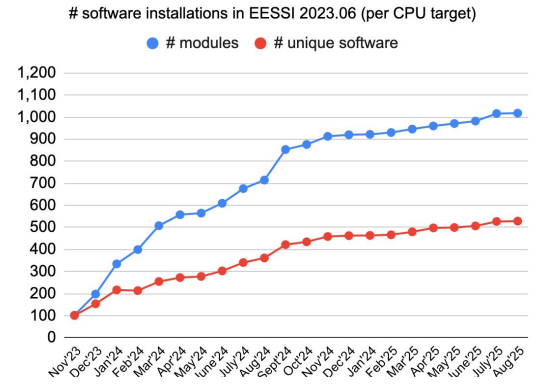
EESSI production repository

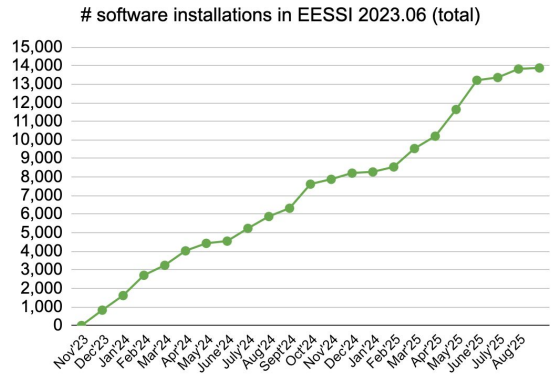
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version 2023.06 is being populated with software via PRs to software-layer repo + build-and-deploy bot

- 13 fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- **Intel Cascade Lake + Ice Lake complete since mid May 2025**
- **Work-in-progress:**
  - **Arm A64FX (for EuroHPC system Deucalion): ~56% done**
- Initial support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 1027 software installations per CPU target**
  (+45 since June'25)
  - **533 different software projects** (+26 since June'25)
    13,902 software installations (across 13+1 CPU targets, +541 since Jun'25)
- Current focus:
  - Adding more GPU (CUDA) software installations
  - Processing incoming contributions
  - Complete set of installations for A64FX
  - Software layer for new EESSI version 2025.06 (see next slide)





EESSI production repository

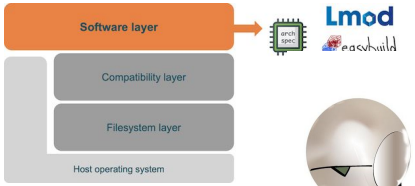
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version 2025.06 is the new EESSI version

- Compatibility layer is in place for `x86_64, aarch64, riscv64`
- Init scripts and modules are available
- Build container based on Debian 12 and `unionfs-fuse` will be used for building software
- The same set of CPU targets will be supported as for 2023.06
- Working towards making `foss/2024a` and `foss/2025a` available
  - GCC compilers (13.3.0 and 14.2.0) are already available
  - Encountered some issues with other packages (`OpenMPI, BLIS, Python`)

Bot for building + deploying software (1/2)



https://github.com/EESSI/eessi-bot-software-layer

- July-August'25
  - PR #312, PR #329, PR #331 rework definition of build targets and user interface to submit jobs **(BREAKING CHANGE)**
    - from `arch_target_map` to `node_type_map`
    - from `architecture:x86_64/amd/zen4` to `for:cpu=x86_64/amd/zen4` [& `on:cpu=x86_64/amd/zen4`]
    - from `accelerator:nvidia/cc80` to `for:gpu=nvidia/cc80` [& `on:gpu=nvidia/cc80`]
    - enable using `ssh` (keys) for obtaining PR contents (supporting private repositories)
  - PR #327, PR #335 support for template values in config setting for scontrol command
  - PR #334 additional option (`last_build`) for only printing the status of the last build (`bot:status` command)
  - PR #340 update action versions in scorecards CI
  - PR #336, PR #339, PR #341 release v0.9.0 **(includes BREAKING CHANGES, see above)**
  - PR #334 take builds for different EESSI versions into account when printing last build status
- Open PRs: change filter logic, support different tasks & fix out of range error

[Bob, Caspar, Kenneth, Lara, Pedro, Sam, Thomas]

Bot for building + deploying software (2/2)

https://github.com/EESSI/filesystem-layer

- PR 232 bundling staging PRs
  - extensively tested
  - several issues & shortcomings documented
  - however, merged and put into production
  - new method needs more work to address shortcomings / missing features / suggested improvements

[Bob, Thomas]

EESSI documentation

eessi.io/docs - GitHub repo github.com/EESSI/docs

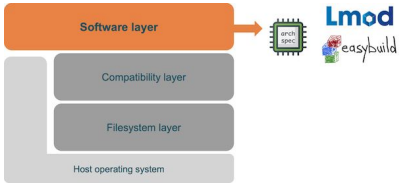
Improvements to the EESSI documentation: ~23 merged PRs (+ 7 for software overview)

Highlights:

- EESSI Events & Training
  - Updated the section in the docs
  - Added EESSI Happy Hour (+ added links to recordings)
  - Added EESSI CI/CD hackathon in Kopenhagen
- Blog posts
  - Work on ROCm (https://www.eessi.io/docs/blog/2025/08/07/rocm/)
- Updating documentation on Software-layer and software-layer-script repos and the bots
- Documentation on GPU support (added mapping of combinations)
- Added EESSI governance
- Added new systems where EESSI is available
- Plus other minor changes to the documentation

[Lara, Pedro, Helena, Bob, Xin, Kenneth, Caspar…]

EESSI test suite



eessi.io/docs/test-suite - github.com/EESSI/test-suite

ReFrame

Highlights

- Add test for numpy (#276)
- skip tests which require internet access on offline machines (#279)
- Various smaller fixes/changes

[Caspar, Sam, Satish, Kenneth, Lara]

Improving NVIDIA GPU support

- Current aim is to support 3 generations of NVIDIA GPUs (Volta, Ampere, Hopper)
  - Results in 39 CPU-GPU combo targets (for 6 of these: native builds on GPU nodes)
  - See https://eessi.io/docs/software_layer/gpu_targets
- Leveraging CUDA sanity check in EasyBuild
  - Currently rebuilding GPU software for all CPU-GPU targets: UCX+UCC+NCCL+OSU-Micro-Benchmarks (#1165), GROMACS (#1166), LAMMPS (#1167), ESPResSo (#1168), LightGBM (#1169)
  - Evaluating results of CUDA sanity check + what's necessary if it fails
  - Also hitting genuine bugs with CUDA on aarch64 (cfr. LAMMPS/Kokkos issue)

[Caspar, Lara, Kenneth]

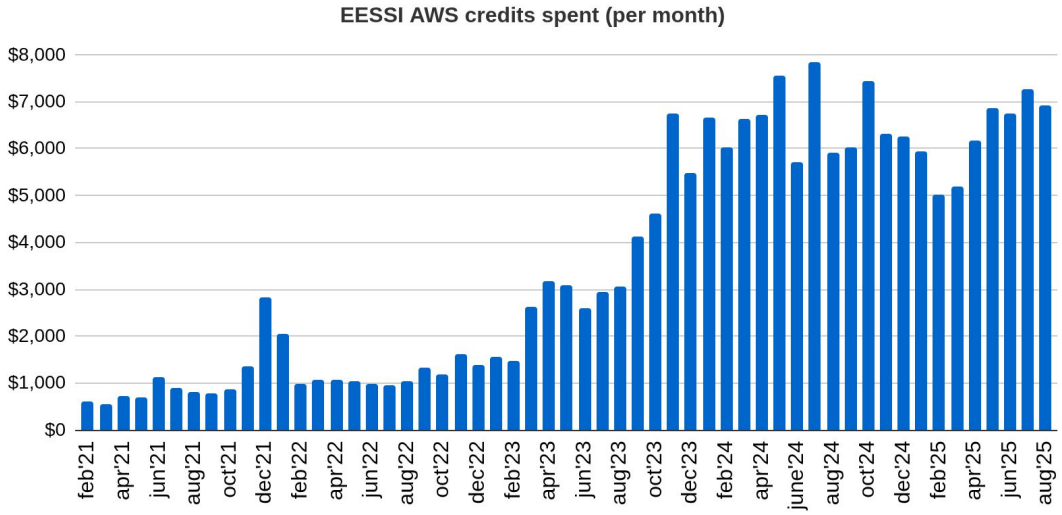
Progress on support for AMD GPUs (ROCm)

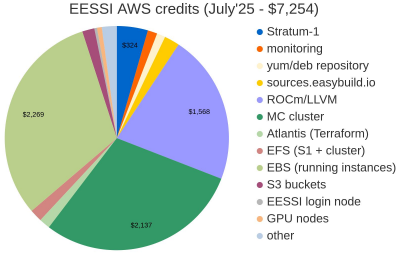
- Work on adding support for AMD GPUs is underway
- Built on the back of much better LLVM support in EasyBuild
- Several PRs open in EasyBuild related to this, still work-in-progress
- Status updates covered in recent EESSI blog post (7 Aug'25):
  - ROCm Integration Progress: Core Components Ready for EasyBuild Release
- Effort was funded by Microsoft Azure to let INUITS contribute to this
- **Can build ROCm 6.4.x on top of EESSI + run the ROCm validation suite**
- Deployment will be done via a dev.eessi.io repository initially to figure out structure, …

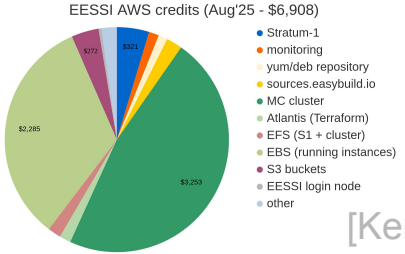
[Bob, Alan, Kenneth, …]

Sponsored AWS credits

- ~$7,254 + $6,908 “spent” in Jul'25 + Aug'25 on Stratum-1 servers, monitoring, demos, sources.easybuild.io,Slurm clusters (build bot)
- **Bulk of consumed credits due to EESSI build-and-deploy bot**
- Regular sync meetings with AWS
- Current credits extended until Dec’25, we have informal confirmation on more sponsored credits for 2026



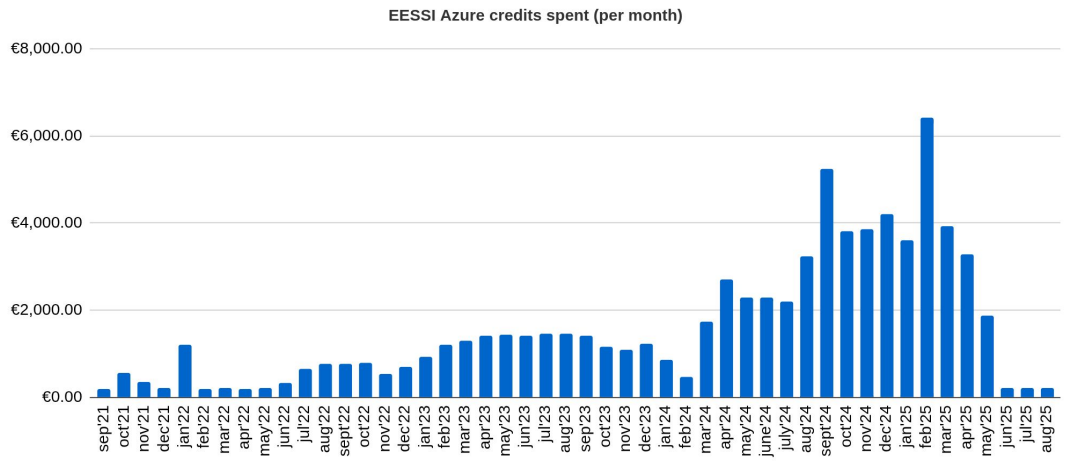




[Kenneth, Bob, Alan]

Sponsored Azure credits

- In July+Aug'25: €213+€211 worth of credits spent
- Used for: Stratum-1 mirror server
  - **Slurm build cluster for dev.eessi.io has been moved to AWS**
  - Costs of keeping CernVM-FS Stratum-1 mirror server up are OK until end of 2025 (via SURF)
- Discussions to extend sponsored credits have stalled…



[Kenneth, Bob, Alan]

EESSI Governance - (interim) Steering Committee

- (interim) Steering Committee
  - Kenneth/Lara, Alan/Davide, Bob/Henk-Jan, Caspar/Satish, Thomas/Terje
- Most recent meetings were held on 4 + 25 August 2025
  - Next one scheduled for 16 Oct 2025
- **Governance for EESSI has been formalized in the documentation**
  - https://www.eessi.io/docs/governance
  - *Feedback is welcome* (e.g. in #governance Slack channel)!
  - Including Terms of Use and Code of Conduct for EESSI
- Discussed EuroHPC call for Transversal Centres of Excellence
  - https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-JU-EUROHPC-2026-COE-LH-01-02

EESSI CI/CD hackathon after EuroHPC User Days

- Thu 2 Oct 2025, 09:00-17:00 CEST
- In the wake of **EuroHPC User Days 2025** (30 Sept + 1 Oct in Copenhagen)
- Being organised in collaboration with EuroHPC (CASTIEL2 and NCC Denmark)
- Hybrid event, max. 25 in-person attendees + virtual attendance via Zoom
- **More information via https://eessi.io/docs/training/2025/hackathon-eurohpc-user-days**
- Currently 13+4 in-person registrations + 20 virtual
  - **We still have a couple of in-person seats left!**
- Free swag for all in-person attendees…
- Raspberry Pi 5 start kit prize for most impressive project! *(sponsored by Do IT Now)*

[Kenneth, Alan, Thomas, Lara]

EESSI Happy Hour sessions

A main theme across several sessions. Each session will include:

- Quick updates: Recent EESSI developments + highlights of upcoming events

- Spotlight topic (~15 min)

- A focused introduction to the day's topic
- May feature:
  - A live demo
  - A guest speaker sharing insights

- Open discussion

- Time for questions, ideas, and community sharing

Started mid August 2025 on Mondays 14:00-15:00 CEST

https://www.eessi.io/docs/training-events/2025/happy-hours-sessions

[Richard]

Next EESSI update meetings

- EESSI update meetings are **bi-monthly**
- First Thursday of the month at 14:00 CE(S)T, only in odd months
  - iCalendar URL for calendar integration
- Next meetings:
  - Thu 6 Nov 2025 14:00 CET (13:00 UTC)
  - Thu **8** Jan 2026 14:00 CET (13:00 UTC)
  - Thu 4 Mar 2026 14:00 CET (13:00 UTC)